Hand Writing 4.1, 4.3, 5.5, 5.6, 5.20

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4.1 Answer each of the following:

- **a.** Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2.
- **b.** Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- **c.** Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2.

a. $R^2 = \frac{SSR}{SST} = 375.47 \div (\underbrace{7825.94 - 20 \times 19.21^2}_{5380.482})$

$= 375.47 \div 445.458 \doteqdot 0.8429$

b. $\hat{\sigma}^2 = \frac{SSE}{N-2}$, $R^2 = 1 - \frac{SSE}{SST} \Rightarrow SSE = (1 - R^2) SST$

$= 0.2089 \times 725.94$

$= 151.648866$

$= \frac{151.648866}{20 - 2} \doteqdot 8.4249$

c. $R^2 = \frac{SSR}{SST} = 1 - \frac{\sum \hat{e}_i^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{182.85}{631.63} \doteqdot 0.7105$

4.3 We have five observations on x and y. They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^{5} \hat{e}_i^2 = 3.6$, $\sum_{i=1}^{5} (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^{5} (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- **a.** the predicted value of y for $x_0 = 4$.
- **b.** the se(f) corresponding to part (a).
- **c.** a 95% prediction interval for y given $x_0 = 4$.
- **d.** a 99% prediction interval for y given $x_0 = 4$.
- **e.** a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

a. $\hat{y}_0 = 1.2 + 0.8 \times 4 = 4.4$ #

b. $var(f) = \hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]$

$\hat{\sigma}^2 = \frac{3.6}{5-2} = 1.2$ $\quad \bar{x} = (3+2+1-1)/5 = 1$

$= 1.2 \left[1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right] = 2.52$ #

$$se(f) = \sqrt{var(f)} \doteq 1.5875$$

c. $t_{3,0.975} = 3.182$

$$\hat{y}_0 \pm t_{3,0.975} \times se(f)$$

$$= 4.4 \pm 3.182 \times 1.5875 \doteq 4.4 \pm 5.0514$$

P.I. $= [-0.6514, 9.4514]$ #

d. $t_{3,0.995} = 5.841$

$$\hat{y}_0 \pm t_{3,0.995} \times se(f)$$

$$= 4.4 \pm 5.841 \times 1.5875 \doteq 4.4 \pm 9.2726$$

P.I. $= [-4.8726, 13.6726]$ #

e. when $x = \bar{x} = 1$

$$\hat{y} = 1.2 + 0.8 \cdot 1 = 2$$

$$var(f) = 1.2\left[1 + \frac{1}{5} + 0\right] = 1.44$$

$$se(f) = 1.2$$

P.I: $2 \pm 3.182 \times 1.2 \doteq 2 \pm 3.8184$

$$= [-1.8184, 5.8184]$$

The width of this interval is much smaller than what we have calculated in (c), as we had learnt in the textbook.

Ch 5. 5.5

a. $Var(b_2|X) = \dfrac{\sigma^2}{\sum_{t=1}^{T}(x_i-\bar{x})^2} = \dfrac{\sigma^2}{\sum x_t^2 - \dfrac{(\sum x_t)^2}{T}}$

$\because \sum x_t^2 - \dfrac{(\sum x_t)^2}{T} \xrightarrow{x_t = t,\ t=1,\dots T} \sum t^2 - \dfrac{(\sum t)^2}{T}$

$= \dfrac{T(T+1)(2T+1)}{6} - \underbrace{\left(\dfrac{T(T+1)}{2}\right)^2 \times \dfrac{1}{T}}$ $\quad \dfrac{T(T+1)^2}{4}$

$= \dfrac{T^3 - T}{12}$

$\Rightarrow Var(b_2|X) = \dfrac{12\sigma^2}{T^3-T}$ and $\lim_{t\to\infty} \dfrac{12\sigma^2}{T^3-T} = 0$

$\therefore$ The estimator is consistent #

b. $Var(\hat{\beta}_2|X) = Var(b_2|X) = \dfrac{\sigma^2}{\sum_{t=1}^{T}(x_t-\bar{x})^2} = \dfrac{\sigma^2}{\sum(0.5^t-\bar{x})^2}$

$\sum(0.5^t - \bar{x})^2 = \sum 0.25^t - 2\times\bar{x}\times\bar{x}\times T + T\times\bar{x}^2$

$= \sum 0.25^t - T\bar{x}^2 = \sum 0.25^t - \dfrac{1}{T}(\sum 0.5^t)^2$

$= \dfrac{0.25(1-0.25^T)}{1-0.25} - \dfrac{1}{T} \times \dfrac{0.5(1-0.5^T)}{1-0.5}$

$= \dfrac{1}{3}(1-0.25^T) - \dfrac{1}{T}(1-0.5^T)^2$

$\Rightarrow Var(b_2|X) = \dfrac{\sigma^2}{\frac{1}{3}(1-0.25^T) - \frac{1}{T}(1-0.5^T)^2}$

$\lim_{t\to\infty} Var(b_2|X) = \dfrac{\sigma^2}{\frac{1}{3} - 0} = 3\sigma^2 \neq 0 \Rightarrow$ Not consistent.

c. In a., increasing T gives more information about β_2 and relationship between x and y. However in b., $x_t = 0.5^T$ becomes 0 for moderate size of T, and more T does not give more information about β_2

5.6 $N=63$ $df=60$ $t_{60;\frac{5\%}{2}}=2$

a. $H_0: \beta_2=0$ $H_1: \beta_2 \neq 0$

$$t = \frac{b_2 - 0}{se(b_2)} \sim t(60)$$

$$t^* \frac{3-0}{\sqrt{4}} = 1.5$$

$1.5 < 2$

At 5% significance level, we cannot reject H_0 $(\beta_2=0)$

because $t^*=1.5<2$ #

b. $H_0: \beta_1 + 2\beta_2 = 5$ $H_1: \beta_1 + 2\beta_2 \neq 5$

$L = b_1 + 2b_2 = 2 + 2\times 3 = 8$

$Var(L) = Var(b_1) + 2^2 Var(b_2) + 2\times 1\times 2\times Cov(b_1, b_2)$

$= 3 + 4\times 4 + 4\times(-2) = 11$

$se(L) = \sqrt{11} \doteq 3.3166$

$$t^* = \frac{8-5}{3.3166} \doteq 0.9045 < 2$$

At 5% significance level, we cannot reject H_0 that $\beta_1 + 2\beta_2 = 5$ because $t = 0.9045 < 2$.

c. $H_0: \beta_1 - \beta_2 + \beta_3 = 4 \quad H_1: \beta_1 - \beta_2 + \beta_3 \neq 4$

$$L = b_1 - b_2 + b_3 = -2$$

$$t = \frac{L-4}{se(L)}$$

$$a = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \qquad Var(L) = a^T \Sigma a = [6, -6, 4] \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$= 16$$

$$se(L) = 4$$

$$t^* = \frac{-2-4}{4} = -1.5$$

$|-1.5| < 2 \Rightarrow$ At 5% significance level,

there's no sufficient evidence to reject H_0

$(\beta_1 - \beta_2 + \beta_3 = 4)$

5.20

a. $\bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1$

$\bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2$

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

$$E(\hat{\beta}_{2,mean} | X) = \beta_2 + \frac{E(\bar{e}_2 | x) - E(\bar{e}_1 | x)}{\bar{x}_2 - \bar{x}_1} = \beta_2$$

$$E(\hat{\beta}_{2,mean}) = E(E(\hat{\beta}_{2,mean} | x)) = E(\beta_2) = \beta_2$$

$\therefore \hat{\beta}_{2,mean}$ is an unbiased estimator of β_2

b. $Var(\hat{\beta}_{2,mean}|x) = Var\left(\frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \Big| x\right)$

$= \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} Var(\bar{e}_2 - \bar{e}_1 | x)$

$Var(\bar{e}_1 | x) = Var\left(\frac{1}{3}\sum_{i=1}^{3} e_i \Big| x\right) = \frac{1}{3^2}\sum Var(e_i|x)$

$= \frac{1}{9} \times 3 \times \sigma^2 = \frac{\sigma^2}{3}$

$Var(\bar{e}_2 | x) = \frac{\sigma^3}{3}$

$\Rightarrow Var(\bar{e}_2 - \bar{e}_1 | x) = \frac{\sigma^2}{3} + \frac{\sigma^2}{3} = \frac{2}{3}\sigma^2$

$\Rightarrow Var(\hat{\beta}_{2,mean} | x) = \frac{\frac{2}{3}\sigma^2}{(\bar{x}_2 - \bar{x}_1)^2}$ #

c. $(\hat{\beta}_{2,mean}) = E[Var(\hat{\beta}_{2,mean} | x)] + Var(E(\hat{\beta}_{2,mean}|x))$

$$\begin{cases} Var(E(\hat{\beta}_{2,mean}|x)) = Var(\beta_2) = 0 \\ E(Var(\hat{\beta}_{2,mean}|x)) = \frac{2}{3}\sigma^2 E\left(\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right) \end{cases}$$

代入： $Var(\hat{\beta}_{2,mean}) = \frac{2}{3}\sigma^2 E\left(\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right)$

d. $Var(\hat{\beta}_{2,mean}) = \frac{4}{N}\sigma^2 E\left(\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right)$

$\lim_{N \to \infty} \frac{4}{N}\sigma^2 E\left(\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right) = 0$ #

e. f $\Rightarrow$ code

g. $E[(s_x^2)^{-1}] \xrightarrow{p} 0.12$

$\because x_i \sim U(0, 10)$ iid, $Var(x) = \frac{(10-0)^2}{12} = \frac{100}{12}$

By Law of Large Number, $s_x^2 = \frac{1}{N-1}\left(\sum_{i=1}^{N}(x_i - \bar{x})^2\right) \xrightarrow{p} Var(x) = \frac{100}{12}$

$\because S_x^2 \xrightarrow{P} \frac{100}{12} \Rightarrow S_x^{-2} \xrightarrow{P} \left(\frac{100}{12}\right)^{-1} = 0.12$

$\Rightarrow E(S_x^{-2}) \xrightarrow{P} 0.12$

② $\frac{4}{(\bar{X}_2 - \bar{X}_1)^2} \xrightarrow{P} 0.16$

$\frac{5+0}{2}$

The bottom half is $U(0,5)$ with mean 2.5

top half is $U(5,10)$ with mean $= \frac{5+10}{2} = 7.5$

Thus when $N \to \infty$, $\bar{X}_1 \xrightarrow{P} 2.5$, $\bar{X}_2 \xrightarrow{P} 7.5$

$\Rightarrow \bar{X}_2 - \bar{X}_1 \xrightarrow{P} 5$

$\frac{4}{(\bar{X}_2 - \bar{X}_1)^2} \xrightarrow{P} \frac{4}{5^2} = 0.16$, Thus $E\left(\frac{4}{(\bar{X}_2 - \bar{X}_1)^2}\right) = 0.16$ #

(h) $\hat{\beta}_{2,mean}$ will not be constant

$\because$ For $\bar{X}_1, \bar{X}_2$, both of them converge into 5 and random samples $\Rightarrow \bar{X}_2 - \bar{X}_1 = 0$

$\therefore Var(\hat{\beta}_{2,mean} | X) = \frac{4\sigma^2}{N(\bar{X}_2 - \bar{X}_1)^2}$ is not able to confirm to converge to 0